

# Hyojung Han

## AI Systems Engineer

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### SUMMARY

- AI systems engineer with 19 years of continuous experience (2007-present) spanning computer vision, recommendation systems, generative AI, and enterprise AI platforms.
- Currently leads product strategy and architecture for three AI platforms at ThakiCloud: inference, training, and enterprise agent automation.
- Founded and ran a generative AI content platform that reached 50K paying users and KRW 400M in cumulative revenue across Japan, Taiwan, and Hong Kong.
- Runs a trilingual technical blog (Korean, English, Arabic), having written 2,260 posts since May 2024 with 608 currently public, backed by 32 working independent R&D implementations.

### CORE TECHNICAL SKILLS

**Agent Harness & Orchestration** Harness Engineering · Loop Engineering · Graph Engineering · MCP Tool Integration · Skill Routing / RAG · LangGraph · Human-in-the-Loop Approval · Policy & Audit Trail · Deterministic Verification Gates · Multi-Agent Consensus

**Training, Fine-tuning & Distillation** SFT · CPT · DPO · GRPO · GKD · LoRA / QLoRA · Knowledge Distillation · FSDP / DDP · TRL · Unsloth · MLflow · Eval Sets & Regression Testing

**Inference, Serving & Compression** vLLM · NVIDIA Dynamo · LMCache · Prefill-Decode Disaggregation · NVFP4 / W4A16 · LLM Compressor · MoE Compression / Pruning · Scale-to-Zero · KServe · Model Routing

**Speech, Vision & Generative Models** VoxCPM2 TTS · Qwen3-TTS · Qwen3-ASR · Speaker Diarization (pyannote) · VLM Video Understanding · Stable Diffusion · DreamBooth · GPT Image · Text-to-Video · EmbeddingGemma · AdaFace / ArcFace · OpenCV / ViT

**Platform & GPU Infrastructure** Kubernetes · Kueue / MultiKueue · ArgoCD / GitOps · Helm · Keycloak · SeaweedFS · CloudNativePG · NATS · Prometheus / Grafana · CUDA · Docker

**Languages & Frameworks** Python · Go (Fiber) · TypeScript · C++ · React 19 · Next.js · FastAPI · PyTorch · PostgreSQL · Redis

**Classical ML & Optimization** LightGBM · CatBoost · XGBoost · scikit-learn · Google OR-Tools · Airflow · Linear Programming / Constraint Optimization · Time Series / Ranking

### PROFESSIONAL EXPERIENCE

**ThakiCloud AI Platform Strategy, Architecture & Team Lead** Seoul, South Korea Apr 2025 ~ Present  
Leads strategy, architecture, and delivery across three AI platforms, and operates the company's multilingual technical blog pipeline.

#### Led strategy and architecture for three AI platforms

- Set product direction and designed the core architecture for inference, training, and agent automation, and led the engineering team by setting experiment priorities
- Directed the design of GPU scheduling, multi-tenancy, and GitOps deployment on Kubernetes, overseeing the build

#### Agent builder integration with internal systems (hands-on)

- Personally designed and built the path that wires the agent builder into internal systems so real work can be assembled and executed as agents
- Owned model quantization (NVFP4, W4A16) end to end, qualifying each build by whether it actually serves before it ships

**2i Studio (Founder) Founder** Yongin, South Korea Mar 2024 ~ Apr 2025  
Founded and ran a generative AI content platform end to end, from product to engineering to operations.

#### Built a face-based generative AI content platform

- Designed an identity-preserving generation pipeline on DreamBooth fine-tuning, growing to KRW 400M cumulative revenue and 50K paying users
- Shipped the full stack (FastAPI/RabbitMQ generation queue, Expo/Next.js app and web) and launched in Japan, Taiwan, and Hong Kong

## **Toss** Senior Machine Learning Engineer Seoul, South Korea

Jan 2021 ~ Feb 2024

Designed and shipped machine learning systems across CDP personalization, generative AI tooling, face authentication, and workforce scheduling.

### **Customer data platform and personalization prediction**

- Built per-service, per-user CTR/CVR prediction models, replacing manual SQL segment queries with automated probability scores
- Designed a separate early-stopping model for A/B tests, cutting test duration by 75%

### **Tosst, a generative graphic design tool for designers**

- Built a Stable Diffusion prompt engine with Toss style templates, cutting character asset production from weeks to 2-3 days
- Saved tens of millions of KRW in outsourcing costs; adopted across Toss blog, web, and app graphics

## **Samsung Electronics** Senior Software Engineer, ML Researcher, Development Manager Suwon, South Korea

Sep 2011 ~ Dec 2020

Built an ML platform, 5G network automation, and a music recognition engine, while also shaping the organization's technical direction.

### **Built an internal cloud machine learning platform**

- Designed a Kubernetes-based ML platform, cutting model deployment lead time from days to hours
- Auto-deployed WaveNet, DeepVoice2, SRGAN, CycleGAN, and SSD models, moving multiple teams' PoCs into production

### **5G vRAN GPU automation and statistical analysis**

- Optimized CUDA-based parallel processing and designed a real-time statistics system, auto-detecting bottlenecks during an early 5G launch
- Ran technical workshops directly with NVIDIA HQ, serving as the point of contact for global collaboration

## **Daum** Machine Learning Engineer Jeju, South Korea

Sep 2008 ~ Aug 2011

Started in computer vision, building object recognition and image search systems.

### **Object recognition and image search system**

- Designed a real-time image matching system at 7M-image scale using histogram-based Hamming embedding and LIS indexing
- Built image hash deduplication, face detection, and category classifiers, applied to representative-image recommendation in shopping search

## **Vision Startup** Software Engineer Seoul, South Korea

Jan 2007 ~ Aug 2008

Began my career in image processing and pattern recognition, the groundwork for later image search systems.

### **Image processing and pattern recognition modules**

- Implemented C++ vision modules that extracted and matched features from camera input
- Iterated on preprocessing and recognition accuracy, building the computer vision fundamentals used later at Daum

## **SYSTEMS BUILT INDEPENDENTLY**

Extends long-held domains, computer vision, signal processing, combinatorial optimization, and recommendation, through agent-driven development. Knowing the field is what makes the speed possible: I know which measurement settles the question, so a paper becomes a measured implementation rather than a summary. Owns and operates 159 repositories, 42 of them completed implementations with measured results.

### **Agents & Automation Harness**

- Document automation that lets staff review only exceptions instead of manually keying quotes and purchase orders
- AI support agent that handles repetitive inquiries so staff focus on real conversations

### **Data & Recommendation**

- Unified scattered department spreadsheets into a standard ETL pipeline, cutting executive status checks from half a day to instant
- Demand-forecast-driven inventory optimization that reduces stockouts and dead stock together

### **Vision**

- 24/7 manufacturing vision inspection AI that unifies judgment criteria that used to vary by inspector
- Self-correcting research-tool automation loop that finds and fixes its own image segmentation errors

## **EDUCATION**

**Yonsei University** M.S. in Computer Science

2007